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Research paper

A task-oriented few-shot spectroscopy benchmark: Empirical evaluation of meta-learning approaches

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ABSTRACT

Near- and mid-infrared (NIR/MIR) spectroscopy, crucial for rapid, noninvasive chemical analysis in industrial applications, is increasingly leveraged by machine learning techniques to extract meaningful information from complex spectral data. Nevertheless, the constraints imposed by limited sample sizes and the occurrence of domain shifts present substantial obstacles to the effectiveness of both deep learning and transfer learning methodologies. In this work, we introduce a unified, task-oriented benchmark that consolidates nine open-access spectral datasets, providing a consistent preprocessing pipeline and clearly defined few-shot tasks. The benchmark is segmented into multiple tests exhibiting varying degrees of heterogeneity, thereby facilitating the evaluation of diverse methodologies across a range of contexts characterized by varying levels of complexity. To the best of our knowledge, meta-learning has not previously been applied in the context of spectrometric analysis. In this study, we confirm that it offers superior performance to traditional transfer learning approaches. A systematic comparison of transfer learning and meta-learning techniques reveals that meta-learning consistently outperforms transfer learning—achieving lower errors in both heterogeneous and homogeneous tasks, consistent with our latent-space analysis showing a more structured cross-task embedding geometry that helps explain these gains. Furthermore, we empirically demonstrate that this benchmark can serve as a starting point for pre-training models on new industrial problems, accelerating adaptation and improving final performance using meta-learning. We complement our analysis with established, model-agnostic interpretability tools, enhancing transparency and industrial relevance in regulated domains.

1. Introduction

The use of machine learning in laboratories has garnered increasing interest due to advances in the field of machine learning in general, and deep learning in particular. Specifically, one of the areas where these techniques are being most widely applied is in the analysis of Near-Infrared Spectroscopy (NIR) and Mid-Infrared Spectroscopy (MIR) data (Zhang et al., 2022b). Spectroscopy is a non-invasive technique that provides detailed information about the chemical composition and physical properties of samples. This technique is extensively used in various industries, such as pharmaceuticals (Li et al., 2020), food (Liu et al., 2022; Olisah et al., 2024), and agriculture (Shen et al., 2022; Ahmadi et al., 2021), mainly due to its rapid and precise analytical capabilities.

Traditionally, regression using Partial Least-Squares (PLS) (Tiecher et al., 2021) has been the most prominent approach for predicting properties or characteristics of samples from spectral data. This technique effectively deals with high-dimensional inputs by modeling latent spaces correlated with the target variables. However, recent advances in Deep Learning (DL) have started to shift this landscape, with neural networks emerging as the most promising solution (Cui and Fearn, 2018; Malek et al., 2018; Ng et al., 2019; Yang et al., 2020; Liang et al., 2024; Yang et al., 2025).

Despite these advances, certain challenges persist and limit the broader adoption of DL in spectroscopic applications. A commonly cited issue is the scarcity of comprehensive datasets (Liu et al., 2022), compounded by the logistical and economic constraints of acquiring large numbers of diverse samples. Moreover, when models are deployed to new scenarios — such as different production years, geographical regions, or variations in measurement conditions — they frequently exhibit significant performance degradation (Shen et al., 2022; Mishra and Passos, 2021; Zhang et al., 2024). Transfer Learning (TL) (Zhuang

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et al., 2020) addresses this challenge by transferring knowledge across domains, reducing the data needed for new tasks.

While advanced Machine Learning (ML) approaches, including TL, have gained some traction in spectroscopic data analysis (Li et al., 2020; Shen et al., 2022; Mishra and Passos, 2021; Zhang et al., 2024), meta-learning remains largely unexplored in this field. Meta-learning, often referred to as "learning to learn", focuses on acquiring generalizable knowledge from a distribution of tasks, enabling rapid adaptation to new tasks with minimal additional data (Hospedales et al., 2021; Huisman et al., 2021). This is particularly appealing in spectroscopic settings, where datasets are often small or costly to obtain. However, a key obstacle to rigorously testing and comparing both TL and meta-learning in spectroscopic applications is the lack of a unified, large-scale benchmark. The absence of a unified benchmark has led to a fragmented landscape, where domain-specific datasets with varying formats and preprocessing pipelines hinder comparability and reproducibility. This fragmentation makes it difficult to assess and compare new contributions, limiting progress in the field under realistic spectroscopic conditions.

To bridge this gap, this work introduces a new benchmark dataset that consolidates multiple NIR and MIR spectral libraries across different application domains. This dataset includes — but is not limited to — soil, fruit, food, and pharmaceutical spectral data. By merging several well-known smaller collections into one coherent resource, our benchmark provides broader geographic coverage, more diverse sample types, and a consistent preprocessing pipeline. We envisage that this unified benchmark will facilitate more robust and transparent comparisons of machine learning methods, including DL, TL, and meta-learning, thereby accelerating progress in industrial and scientific spectroscopy. In addition, a methodology is presented that is based on inter-task heterogeneity metrics. This methodology allows for the quantification of task diversity in advance and the estimation of the potential effectiveness of meta-learning approaches prior to conducting full — and often computational costly — empirical evaluations.

The remainder of this article is structured as follows. Section 2 positions our study within spectroscopic and machine learning literature. Section 3 details data acquisition, preprocessing, and the meta-learning framework, outlining the proposed methodology to be followed, based on inter-task heterogeneity. Section 4 highlights key experimental findings under different data constraints, analyzing the effectiveness of the proposed methodology. Finally, Section 5 explores the implications of our results, and Section 6 offers concluding remarks and future directions.

2. Related works

NIR and MIR spectroscopy are widely used techniques in laboratories due to their non-destructive, fast, and efficient capability to analyze the physicochemical properties of various types of samples (Zhang et al., 2022b). Traditionally, these techniques have been combined with classical ML methods, such as PLS (Tiecher et al., 2021), Support-Vector Machine (SVM) (Galán-Freyle et al., 2020), and Random Forests (RFs) (Zhang et al., 2020), demonstrating effectiveness in tasks ranging from soil analysis to quality control in food products. However, these approaches often rely on extensive manual preprocessing and face difficulties when confronted with the high dimensionality and inherent noise of spectral data (Ng et al., 2019).

Recent advances in DL have mitigated some of these limitations. In particular, Convolutional Neural Networks (CNNs) have shown an exceptional capability to learn complex representations from raw spectral data, thus reducing or eliminating the need for manual feature engineering (Shen and Viscarra Rossel, 2021; Ng et al., 2019; Haffner et al., 2025). Other neural architectures, such as Recurrent Neural Networks (RNNs) (Yang et al., 2020; Tian et al., 2025; Fu et al., 2025) and Autoencoders (Grossutti et al., 2022; Kong et al., 2024), have likewise been successful in various spectral analysis tasks. Nevertheless, an important drawback of DL-based methods is their strong reliance on large training datasets; performance typically improves with data size (Ng et al., 2020), making data scarcity a critical challenge. A related issue concerns limited generalization to new conditions (e.g., different measurement setups or geographic regions), which can cause significant performance degradation (Mishra and Passos, 2021; Shen et al., 2021).

TL (Zhuang et al., 2020) has emerged as a promising avenue to address these issues by transferring knowledge from a source domain, ideally relying on a sufficiently large and comprehensive dataset, to a target domain with fewer samples. Several studies have demonstrated the utility of TL in spectrometric applications. For instance, Shen et al. (2022) and Padarian et al. (2019) leveraged global soil datasets to build localized predictive models, while Mishra and Passos (2021) applied TL to adapt models for seasonal variations in NIR spectra of mangoes or to account for recipe changes in melamine production. In both cases, TL significantly outperformed traditional approaches, underscoring its potential to boost adaptability and accuracy under shifting conditions.

Beyond supervised TL, self-supervised contrastive methods provide a prominent route to learn general-purpose representations under scarce labels. In this family, *SimCLR* offers a strong and simple baseline that maximizes agreement between two heavily augmented views using an InfoNCE objective, highlighting the importance of augmentation design, a non-linear projection head, and (typically) large-batch training (Chen et al., 2020a). Scaling this recipe, SimCLRv2 shows that larger backbones and semi-supervised fine-tuning substantially improve label efficiency (Chen et al., 2020b). Complementary designs such as MoCo replace large batches with a momentum-updated dictionary to enable competitive contrastive learning with modest hardware (He et al., 2020). On the theoretical side, CPC motivates the InfoNCE objective, while subsequent analyses relate contrastive success to *alignment* and *uniformity* of normalized features and to the choice of informative view pairs (van den Oord et al., 2018; Wang and Isola, 2020; Tian et al., 2020). In the supervised regime, Supervised Contrastive Learning generalizes the contrastive loss to multiple positives per class and can outperform cross-entropy under strong augmentations (Khosla et al., 2020). Similar contrastive pipelines have also been applied to spectroscopy (e.g., Raman) (Li et al., 2022).

While TL and contrastive learning have gained some attention in spectroscopic data analysis, the application of meta-learning remains rare in this domain. However, in other industrial contexts, meta-learning has already been applied successfully, for instance, in adaptive defect inspection in production (Zhang et al., 2023b), in repair decision for aircraft structures (Che et al., 2023), and in food origin detection systems (Chen et al., 2025). Meta-learning, often described as "learning to learn", optimizes the learning process itself, enabling rapid adaptation to new tasks with minimal data (Hospedales et al., 2021; Huisman et al., 2021). Meta-learning strategies are particularly appealing for spectroscopic scenarios where dataset sizes are often constrained, yet samples may vary significantly due to regional, seasonal, or instrumental factors.

Broadly categorized into metric-based, model-based, and optimization-based approaches, meta-learning methods have demonstrated remarkable success learning new tasks with few samples in fields like image classification (e.g., mini-ImageNet (Vinyals et al., 2016) and Omniglot (Lake et al., 2015)). Metric-based methods focus on learning similarity functions that enable comparisons between samples from new tasks and previously seen data, allowing effective generalization to unseen tasks. Well-known examples include Matching Networks (Vinyals et al., 2016) and Prototypical Networks (Snell et al., 2017). Model-based approaches center on designing architectures capable of rapid adaptation by maintaining an internal state or representation that evolves as task data is sequentially processed, as seen in methods like Memory-Augmented Neural Networks (MANNs) (Santoro et al., 2016) and Simple Neural Attentive Meta-Learner (SNAIL) (Mishra

et al., 2018). Finally, optimization-based approaches aim to learn efficient parameter update strategies for new tasks, often by optimizing model initialization for quick adaptation. The MAML algorithm (Finn et al., 2017) is a prominent example, along with Meta-LSTM (Ravi and Larochelle, 2017), which employs a task-specific optimization policy using a learned LSTM-based optimizer. Despite their success in other fields (Kshetrimayum and Huang, 2024; Lu et al., 2023), their potential for complex regression tasks — particularly in spectroscopic contexts — remains largely unexplored.

Moreover, existing meta-learning research frequently focuses on simplified or academic classification tasks, ignoring the more complex challenge of regression in adaptation under few-shot conditions. Moreover, these studies fail to reflect the operational constraints, large domain shifts, and heterogeneous data distributions characteristic of real-world industrial settings. Despite the fact that general-purpose meta-learning benchmarks, including Meta-Dataset (Triantafillou et al., 2020) and Meta Omnium (Bohdal et al., 2023), encompass multiple datasets, with the latter including a variety of tasks (some of which are regression-based), these resources remain centered on computer vision. Furthermore, the regression tasks featured in these resources are largely simplistic, failing to capture the complexities and demands of real-world industrial applications. Consequently, these benchmarks do not effectively address the distinctive challenges associated with industrial spectroscopic data, such as one-dimensional signals characterized by high noise levels and domain-specific preprocessing requirements. In order to address this issue, a consolidated NIR-MIR benchmark spanning diverse industrial and scientific domains is hereby introduced. The demonstration will illustrate how state-of-the-art TL and meta-learning can be utilized in this manner.

3. Materials and methods

This section presents the core contribution of the study: the construction, analysis, and validation of a unified, task-oriented benchmark for evaluating learning methodologies in industrial spectroscopy. In particular, the proposed benchmark includes regression tasks which require predicting precise continuous values. Such tasks are especially challenging under limited data scenarios, as they demand capturing nuanced relationships that classification problems typically do not show.

This section is organized as follows. Section 3.1 describes the benchmark's structure and the data that comprise it. Section 3.2 presents the preprocessing applied. Section 3.3 defines two metrics for measuring the heterogeneity of the benchmark's tasks, which will be essential for the analysis of results. Finally, Section 3.4 presents the methods applied to the benchmark for the experimental comparison.

3.1. Benchmark definition

The main contribution of this work is the creation of a unified benchmark that consolidates spectral dataset from diverse application domains (soil, fruit, fuels, and others) into a common resource. Designed in a task-oriented manner, the benchmark enables fair comparison of different learning strategies while addressing critical challenges in industrial spectroscopy. It also allows us to simulate scenarios involving data scarcity or concept drift that frequently arise in industrial problems.

3.1.1. Task-oriented structure

To enable a flexible evaluation of different industrial spectroscopic problems, we have organized our proposed benchmark in a *task-oriented* manner. Specifically, each dataset and its associated properties has been segmented into multiple tasks, where each task encapsulates a distinct industrial problem (further details can be found in Appendix A). This structure not only yields a large and diverse collection of tasks for training, validation, and testing under varying degrees of domain shift, data scarcity, or noise, but also naturally simulates the emergence of new tasks in real-world settings with limited data. Moreover, the framework's modular design facilitates the gradual enhancement of the benchmark through the systematic addition of new tasks. Researchers are empowered to integrate their own datasets as one or multiple tasks, enabling the continuous expansion and adaptation of the suite of problems. Consequently, the benchmark facilitates systematic experimentation in both TL and meta-learning scenarios.

Each task is further split into two subsets: the *support* subset provides the labeled examples used to tailor the model to the specific task, while the *query* subset, composed of previously unseen samples, measures the model's ability to generalize.

3.1.2. Data collection and test design

To capture the wide range of challenges inherent to industrial spectroscopy, we compiled a large collection of open-access spectral data from multiple repositories. Our approach involved: (i) selecting relevant databases; (ii) harmonizing formats and preprocessing protocols; and (iii) unifying target variables to define consistent tasks. Table 1 lists all the dataset included in our benchmark, accompanied by a concise summary of their characteristics, the manner in which each set has been segmented into tasks, and the support/query division within them. The complete details related to the datasets can be found in Appendix A.

Licensing and reuse. We do not mirror third-party datasets; instead, we provide links and automated preparation scripts so that each source retains its original terms. Most tasks in the benchmark originate from openly licensed sources (CC BY/MIT), which generally allow reuse — including *commercial* use — with proper attribution. A small subset corresponds to datasets hosted by Eigenvector that *do not* declare an explicit license on their landing pages. Although these Eigenvector datasets have been used in the spectroscopy literature, the practical constraint for users is **commercial use**, for which explicit authorization should be sought. To accommodate industrial settings, our repository includes a convenient option to **download only those datasets whose licenses allow commercial use**.

These datasets were ultimately grouped into three distinct categories or tests, which together constitute the benchmark and allow us to investigate the behavior of different methodologies under varying conditions:

- *Mango Test:* The objective of this test is to predict the dry matter percentage (%DM) of mango samples collected in different years and regions. In this test, we compare two partitioning schemes — by year of harvest and by year-region combination (see Appendix A.2) — to evaluate model robustness under different conditions. The test comprises a total of 12 tasks: 4 for partitioning by year and 8 for partitioning by year-region; and a total of 11,691 samples.
- *Soil Test:* This test involves the prediction of different soil properties from samples collected across various regions. There are two versions of this test: one with NIR spectra as input and another with MIR spectra (see Appendix A.1). There are a total of 889 tasks, 185 of which are for the NIR version and 704 of which are for the MIR version. This test includes 88,867 instances.
- *TRIP Test:* The objective of this test is to predict the properties of various datasets. Each task will represent an industrial problem in itself. There will be a single version in which tasks will be drawn from the remaining datasets in Table 1. There are a total of 24 tasks and 11,209 instances.

The *Mango test* comprises tasks with broadly similar feature distributions, exhibiting only minor variations due to natural differences among samples collected in different years or regions. The Soil test introduces additional heterogeneity by combining the prediction of different properties with the natural variability of distinct geographic

Table 1
Overview of the nine spectral datasets used in this study. **Samples**: total number of spectra; **Wavelength Range**: measurement interval; **# spectral bands**: number of sampled wavelengths; **# tasks**: number of tasks generated from each dataset; **Tasks definition**: "X properties/recipes/components" indicates X independent tasks (one per item); Datasets marked with † indicate a random 50% support/query split; all others preserve their original split. **Licensing legend.** OL: open license (CC BY/MIT; reuse including commercial use with attribution). NL: no explicit license stated on the dataset landing page (Eigenvector); redistribution is discouraged and commercial use requires authorization. We do not mirror third-party data and provide links/scripts so that each dataset retains its original terms.

Dataset	Samples	Wavelength range	# spectral band	# Tasks	Tasks definition
†Soil (OSSL)^{OL}(Anon, 2021)	88,867	400–16,667 nm	2752	889	Region × Props
Mango^{OL}(Anderson et al., 2020)	11,691	309–1149 nm	281	12	Season Season × Region
†Diesel Fuel^{NL}(Eigenvector)	784	750–1550 nm	410	7	7 properties
†Corn^{NL}(Eigenvector)	240	1100–2498 nm	700	4	4 properties
CGL^{NL}(Eigenvector)	231	1104–2495 nm	117	4	4 components
NIR Shootout^{NL}(Eigenvector)	654	600–1898 nm	650	3	3 properties
†Melamine^{OL}(Nikzad-Langerodi et al., 2018, 2020) (GitHub)	8117	1433–1803 nm	346	4	4 recipes
Wheat Kernel^{OL}(Figshare)	523	850–1050 nm	100	1	1 property
Egg^{OL}(Ramírez-Morales, 2017)	660	740–1070 nm	331	1	1 property

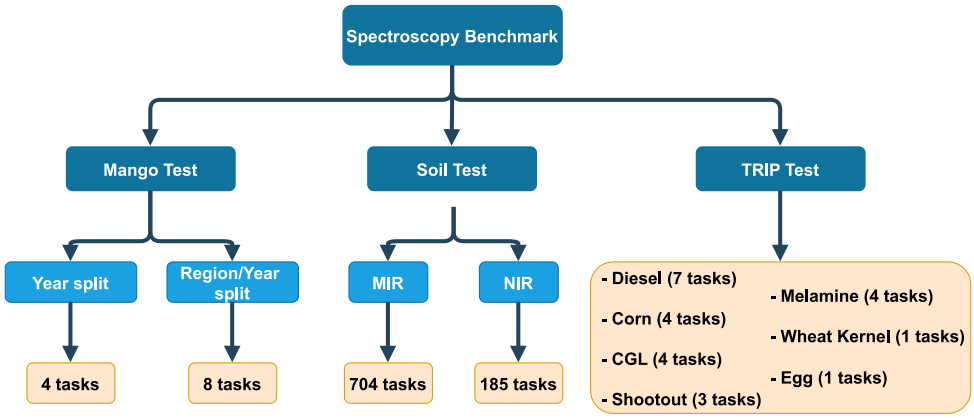


Fig. 1. Structure of the proposed spectroscopy benchmark, divided into three main tests (Mango, Soil, and TRIP). For each test, the partitions of datasets that compose it and the resulting number of tasks are indicated, facilitating the evaluation of different methodologies in varied scenarios.

areas. Finally, the *TRIP test* is the most heterogeneous: each task corresponds to an entirely different problem with non-overlapping characteristics. These three perspectives enable a systematic study of the strengths and weaknesses of the evaluated methodologies across diverse settings.

Fig. 1 illustrates the complete structure of the benchmark, presenting the partitions and the number of tasks assigned to each test.

3.2. Preprocessing and benchmark construction

Integrating multiple datasets requires a unified preprocessing pipeline to guarantee information homogeneity. The adopted procedure comprises:

1. **Smoothing and Differentiation:** A Savitzky–Golay filter is applied to compute the first derivative, thereby reducing noise and accentuating relevant spectral variations. This technique has been widely used in spectroscopy applications as a way to reduce noise and increase predictive performance (Helin et al., 2022; Zhang et al., 2023a; Rinnan et al., 2009; Roger et al., 2022). Figure A.9 shows the effect that this filter has on the spectra and the advantage of its use when there are either noise problems or, more specifically, calibration problems between different instruments.
2. **Target Normalization:** The response variables are standardized in $(\mu, \sigma) = (0, 1)$ to ensure that each task contributes equally during training, preventing those with inherently higher value ranges from disproportionately influencing the gradient updates.

These steps convert the raw data into well-defined tasks suitable for the comparative evaluation of diverse learning strategies.

3.3. Inter-task variability metrics

As introduced in Section 3.1.2, the *Mango* test groups tasks with very similar distributions of characteristics (minor variations by year/region). The *Soil* test introduces heterogeneity by predicting different properties in different regions, and the *TRIP test* brings together tasks that are totally disparate. In this section, we quantify this inter-task variability employing two complementary metrics:

Maximum Mean Discrepancy (MMD). MMD is a kernel-based distance between probability distributions. By mapping the samples of two tasks into a reproducing-kernel Hilbert space (we employ a Gaussian kernel), we compute the squared distance between their mean embeddings. Low MMD values indicate similar feature distributions, whereas high values signal a substantial shift, an idea widely used in domain adaptation and generative-model validation (Gretton et al., 2012).

Feature Importance Similarity (FIS). FIS compares the vectors that summarize the relevance of each input feature for prediction in different tasks. Given an importance vector (Pearson correlation coefficients) for each task, we calculate its cosine similarity with the corresponding vector from every other task. A value close to 1 indicates that both tasks rely on essentially the same spectral bands, revealing underlying homogeneity despite possible marginal distribution differences.

Table 2 confirms the anticipated differences in heterogeneity among benchmark tests: *TRIP* exhibits the highest heterogeneity, *Soil* displays moderate variability due to diverse properties and geographic factors, while *Mango* is clearly the most homogeneous.

Significant differences between source and target domains severely limit the effectiveness of transfer learning (Zhang et al., 2022a; Ehrig

Table 2
Task variability metrics: MMD and FIS for the Mango (Year and Year-Region), Soil (MIR and NIR) and TRIP tests. Low MMD values indicate similar feature distributions between tasks, while high values signal a substantial shift. FIS values close to 1 indicate that the tasks are homogeneous, whereas low values reflect notable differences in feature relevance.

Metric	MMD	FIS
Mango (Year)	0.0320	0.7268
Mango (Year-Region)	0.0356	0.5330
Soil (MIR)	0.1781	0.1942
Soil (NIR)	0.1576	0.1642
TRIP	0.2984	0.0813

et al., 2024), especially when adapting to new tasks with scarce and loosely related data—as exemplified by the *TRIP test* scenario. We aim to demonstrate that meta-learning strategies are better suited to these challenging conditions, outperforming traditional TL approaches.

3.4. Methods dealing with the benchmark

To analyze the performance under limited data availability and inter-task variability, we selected six representative methods:

- Fine-tuning (FT) is a transfer learning strategy in which a pretrained model is fully retrained on limited new-task data, leveraging prior knowledge.
- SNAIL employs temporal convolutions and attention to effectively capture short- and long-range dependencies, enabling quick adaptation (Mishra et al., 2018).
- MAML learns a parameter initialization optimized for quick adaptation through episodic meta-training (Finn et al., 2017).
- Prototypical Networks predict new instances using weighted distances to learned prototypes computed from support set embeddings (Snell et al., 2017; Bohdal et al., 2023).
- SimCLR+MLP combines a contrastive learning pretraining (SimCLR) on augmented spectral pairs (noise, rescale/offset, smoothing, random crop) (Chen et al., 2020a) with a frozen encoder and a two-layer MLP trained on the support during adaptation (regression).
- SimCLR+PR combines a SimCLR-pretrained encoder with a ProtoNet-style adaptation that predicts via weighted distances to support prototypes in the embedding space, without further training (Snell et al., 2017; Zhang et al., 2022d).

The three meta-learning approaches were deliberately chosen to span the canonical families of meta-learning: *metric-based* (Prototypical Networks), *optimization-based* (MAML), and *model-based* (SNAIL).

Comparing these methods offers a comprehensive view of their adaptability and robustness to inter-task variability—key requirements for optimizing industrial spectroscopy applications. Implementation details and the empirical analysis are provided in Sections 4 and 5.

3.5. Qualitative model-agnostic explainability procedures

All compared approaches in this study share a 1D convolutional encoder (ResNet1D) and a scalar regression head, which enables model-agnostic explainability across FT, MAML, Prototypical Networks, and SNAIL. This backbone has previously been shown to be an effective choice for spectroscopic signals (Jiang et al., 2020; Zhang et al., 2022c; ChenDanny et al., 2022). We report two complementary methods that yield wavelength-level overlays for qualitative inspection in spectroscopy regression.

SHAP approximates Shapley values using model gradients to quantify each feature's marginal contribution to an individual prediction; these local attributions can be aggregated to obtain global importance profiles (Lundberg and Lee, 2017).

XGrad-CAM (1D) is an adaptation of gradient-weighted class activation mapping to one-dimensional inputs that forms class-discriminative saliency maps by weighting convolutional feature maps with gradient× activation scores and projecting them onto the input to highlight the most influential regions (Fu et al., 2020).

The wavelength-level overlays are practical decision-support tools for spectroscopists: they surface, at a glance, the regions most influential for each prediction, create a transparent line of sight from model output to input spectrum, and produce an auditable trace that can be archived and revisited. Because they are model-agnostic and consistent across FT, MAML, Prototypical Networks and SNAIL, they drop cleanly into QA/QC and method-development workflows (e.g., pharmaceutical and food safety), helping reviewers focus attention, prioritize follow-ups, and document the rationale for approvals or investigations. In short, they turn otherwise opaque regressors into reviewable instruments that speak the language of spectroscopy — wavelengths, bands, and regions — so domain experts can interrogate, compare, and report with confidence.

4. Results

This section presents the empirical results obtained using the proposed benchmark. In Section 4.1, a quantitative comparison is presented between the few-shot performance of FT, MAML, Prototypical Networks, SNAIL, two contrastive learning baselines (SimCLR+MLP and SimCLR+PR), and a model trained solely on the 25 support examples per task. Section 4.2 explores how each method adapts as the number of support instances is increased, thereby illustrating their scalability with more data. Building on this, Section 4.3 focuses on *extreme few-shot* regimes (5–10 shots). Next, Section 4.4 examines the impact of incorporating diverse meta-training tasks, demonstrating how task heterogeneity can enhance overall performance. In addition, Section 4.5 analyzes the latent-space structure and target–latent alignment, and Section 4.6 evaluates robustness to spectral noise and instrument drift; finally, Section 4.7 presents a qualitative interpretability case study on *Mango*. Overall, the results in this section demonstrate the advantages of meta-learning for few-shot industrial spectroscopy problems. They show that meta-learning outperforms baseline alternatives — such as FT or contrastive learning — both in quantitative metrics and qualitative analyses.

To ensure a fair comparison among all methods, we employed a shared ResNet-based baseline architecture so that any performance differences arise from the methodology itself rather than the model design. Likewise, each method's hyperparameters were tuned independently via grid search to guarantee that every approach operates under its own optimal conditions. Full details of the architecture and hyperparameter configurations can be found in Appendix B, including a learning-rate sensitivity study for the two best-performing meta-learning methods (Appendix B.3).

All code and preprocessing pipelines have been made publicly available,[1] and a script is provided to download the datasets, ensuring full reproducibility and facilitating extensions by the community.

4.1. Few-shot performance analysis

Data scarcity is a common challenge in industrial applications, often impairing model performance. In this section, we present the results of applying various techniques under few-shot conditions. The different models were initially trained using the training tasks from each test described in Section 3.1.2. For the test tasks, models were adapted using only 25 support instances over 10 adaptation epochs

[1] https://github.com/ivgase/spectroscopy-fewshot-benchmark

Table 3
Comparison of performance and percentage improvement of each model relative to Fine-tuning. A critical-difference analysis over these results (see Appendix C) confirms two distinct cliques at $\alpha = 0.05$: Prototypical Networks and MAML form the top group (no significant difference between them), whereas the remaining methods trail behind.

(a) Mean Squared Error (MSE) with % improvement relative to Fine-tuning (FT).

Model	Mango (Year)	Mango (Year-Region)	Soil (MIR)	Soil (NIR)	TRIP
Individual	1.239 (−266.6%)	1.023 (−221.7%)	0.769 (8.6%)	0.860 (−15.4%)	0.759 (74.5%)
SimCLR+MLP	1.049 (−210.5%)	0.888 (−179.2%)	0.927 (−10.2%)	0.828 (−11.2%)	0.780 (73.7%)
SimCLR+PR	0.848 (−150.8%)	0.840 (−164.1%)	0.777 (7.6%)	0.817 (−9.6%)	0.705 (76.3%)
Fine-tuning	0.338 (—)	0.318 (—)	0.841 (—)	0.745 (—)	2.972 (—)
SNAIL	0.342 (−1.2%)	0.444 (−39.6%)	0.736 (12.5%)	0.656 (11.9%)	1.457 (51.0%)
MAML	0.225 (33.4%)	0.263 (17.3%)	0.695 (17.4%)	0.759 (−1.9%)	**0.555** (81.3%)
Prototypical Networks	**0.183** (45.9%)	**0.173** (45.6%)	**0.432** (48.6%)	**0.510** (31.5%)	0.580 (80.5%)

(b) Mean Absolute Error (MAE) with % improvement relative to Fine-tuning (FT).

Model	Mango (Year)	Mango (Year-Region)	Soil (MIR)	Soil (NIR)	TRIP
Individual	0.911 (−100.7%)	0.812 (−82.9%)	0.561 (6.8%)	0.640 (−10.0%)	0.648 (49.7%)
SimCLR+MLP	0.857 (−88.7%)	0.777 (−74.9%)	0.641 (−6.4%)	0.622 (−6.9%)	0.628 (51.3%)
SimCLR+PR	0.688 (−51.5%)	0.669 (−50.8%)	0.653 (−8.4%)	0.671 (−15.3%)	0.596 (53.7%)
Fine-tuning	0.454 (—)	0.444 (—)	0.602 (—)	0.582 (—)	1.288 (—)
SNAIL	0.476 (−4.8%)	0.544 (−22.5%)	0.560 (7.0%)	0.560 (3.8%)	0.992 (23.0%)
MAML	0.382 (15.9%)	0.398 (10.4%)	0.534 (11.3%)	0.585 (−0.5%)	0.494 (61.6%)
Prototypical Networks	**0.345** (24.0%)	**0.319** (28.2%)	**0.351** (41.7%)	**0.424** (27.1%)	**0.490** (62.0%)

(c) Coefficient of Determination (R^2) with % improvement relative to Fine-tuning (FT).

Model	Mango (Year)	Mango (Year-Region)	Soil (MIR)	Soil (NIR)	TRIP
Individual	−0.174 (−125.6%)	−0.015 (−102.2%)	0.288 (43.3%)	0.129 (−47.6%)	0.175 (105.9%)
SimCLR+MLP	0.005 (−99.3%)	0.138 (−79.7%)	0.138 (−31.2%)	0.156 (−36.7%)	0.225 (107.6%)
SimCLR+PR	0.121 (−82.2%)	0.147 (−78.3%)	0.214 (6.6%)	0.165 (−33.1%)	0.291 (109.8%)
Fine-tuning	0.679 (—)	0.678 (—)	0.201 (—)	0.246 (—)	−2.975 (—)
SNAIL	0.660 (−2.8%)	0.516 (−23.9%)	0.207 (3.0%)	−0.992 (−503.3%)	−0.785 (73.6%)
MAML	0.767 (13.0%)	0.732 (8.0%)	0.327 (62.7%)	0.210 (−14.6%)	0.344 (111.6%)
Prototypical Networks	**0.826** (21.6%)	**0.825** (21.7%)	**0.610** (203.5%)	**0.492** (100.0%)	**0.429** (114.4%)

to simulate a few-shot scenario; this number of epochs was chosen to balance sufficient convergence on the limited support set against the risk of overfitting. Performance was subsequently evaluated on the corresponding query instances. Additionally, we included a baseline model that is trained solely on these 25 instances without any prior knowledge transfer. This baseline serves to highlight the advantages of specialized few-shot techniques in such industrial contexts.

Table 3 summarizes the obtained results. Regarding the representation-learning baselines, SimCLR+PR generally outperforms SimCLR+MLP and both remain below the meta-learning methods. On the highly heterogeneous *TRIP* test, both SimCLR variants markedly improve over FT, though they still fall short of Prototypical Networks and MAML. In contrast, on the more homogeneous *Mango* tests, the SimCLR variants lag behind FT, and on *Soil* they are competitive in MSE but do not consistently surpass FT in MAE or R^2. Across all tests, Prototypical Networks consistently achieved the best performance. In the *Mango* test, which is highly homogeneous, the FT technique adapts almost as well as the meta-learning approaches, especially MAML. The *Soil* test occupies an intermediate position: here FT performs reasonably well, but Prototypical Networks and MAML still yield the strongest results. Finally, in the *TRIP test* — with the strongest heterogeneity — conventional FT is markedly undercut, and Prototypical Networks and MAML clearly outperform all other methods. These outcomes are in line with the variability analysis of Section 3.3: task heterogeneity is a decisive factor behind the performance patterns reported in Table 3.

The *TRIP test* appears to be especially favorable for meta-learning approaches, with Prototypical Networks and MAML achieving MSE improvements of approximately 81% over FT and approximately 25% over the individual baseline model. Even in scenarios with low data heterogeneity, where FT accomplishes competitive results, meta-learning approaches such as Prototypical Networks achieve the best performance, proving meta-learning applicability in all kinds of few-shot scenarios.

Overall, the Individual baseline and the two SimCLR variants did not dominate across tests: despite clear gains in the heterogeneous *TRIP* setting, they underperform FT on *Mango* and only match or slightly trail it on *Soil*. This underscores the importance of task-aware knowledge transfer via meta-learning in these industrial scenarios.

Finally, while SNAIL has demonstrated strong performance in classification tasks (Mishra et al., 2018), its adaptation to regression in our study did not yield competitive results.

Additional experiments were conducted by increasing the number of adaptation epochs (see Appendix D). The results remain mostly stable, confirming that the performance is not merely due to using a limited number of epochs, but rather reflects the robustness of the observed trends.

4.2. Performance analysis under increased adaptation data availability

To evaluate the behavior of the competing methods beyond the scarce-data regime, we progressively increased the number of support instances available during task adaptation. The resulting learning curves for all three benchmark tests — *Mango* (Year & Year-Region), *Soil* (MIR & NIR), and *TRIP* — are combined in Fig. 2.

In the *Mango test* (Fig. 2a), FT benefits markedly from each new support example, eventually approaching the plateau reached by meta-learning. SNAIL follows a similar trend — albeit more gradually — and surpasses the individual model once task similarity is sufficiently high. These curves demonstrate that, given enough labeled examples in a homogeneous setting, classical FT can perform similar to meta-learning.

The Soil experiments occupy the middle ground in terms of inter-task heterogeneity (Fig. 2b). In this case, Prototypical Networks achieve the best performance by a significant margin, underscoring the advantage of meta-learning when tasks predict different properties and exhibit intermediate variability.

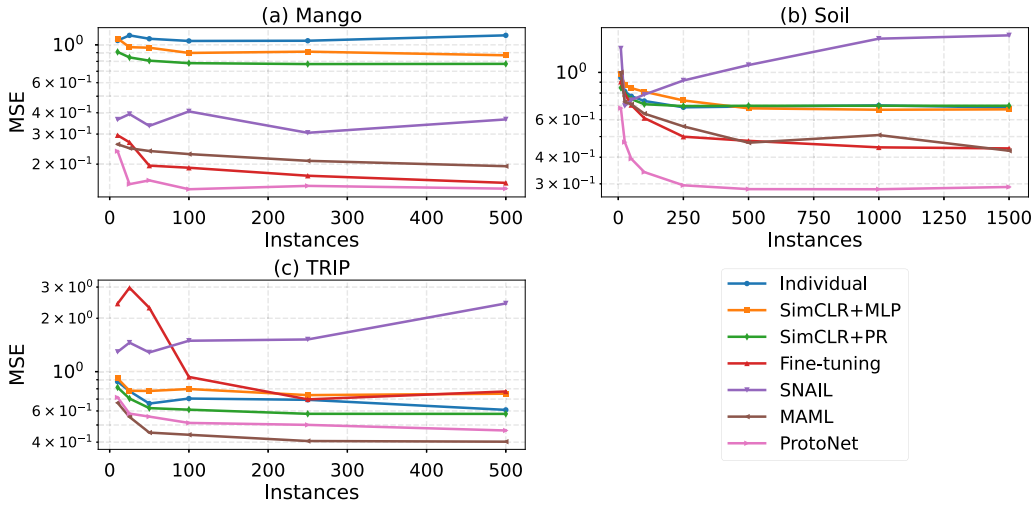


Fig. 2. Performance metrics for the different benchmark tests as the number of instances increases. For clarity, the Soil results (MIR and NIR experiments) and the Mango results (Year and Year-Region experiments) have been aggregated by taking their mean.

Table 4
Coefficient of determination (R^2) under extreme few-shot settings (5 and 10 support instances) and a baseline condition (25 instances), reported for each benchmark and model. For brevity, results for the individual Mango and Soil sub-benchmarks are aggregated. The best value in each column is highlighted in bold.

Model	Mango			Soil			TRIP		
	5-shot	10-shot	25-shot	5-shot	10-shot	25-shot	5-shot	10-shot	25-shot
Individual	−0.049	−0.022	−0.092	0.006	0.076	0.208	−0.310	0.122	0.179
SimCLR+MLP	−0.295	−0.044	0.071	−0.054	0.042	0.147	−0.207	0.073	0.225
SimCLR+PR	−0.112	0.077	0.134	−0.133	0.140	0.189	−0.101	0.175	0.291
Fine-tuning	0.575	0.712	0.736	0.069	0.107	0.223	−2.200	−2.190	−2.975
SNAIL	−1.658	0.416	0.588	−260.705	−113.312	−0.393	−2.268	−0.898	−0.785
MAML	**0.601**	0.744	0.746	0.006	−0.051	0.269	**−0.095**	0.249	0.344
Prot. Networks	0.424	**0.767**	**0.851**	**0.149**	**0.336**	**0.551**	−0.115	**0.251**	**0.429**

In the *TRIP test*, the most heterogeneous collection (Fig. 2c), all three meta-learning algorithms continue to outperform both conventional FT, the individual and SimCLR baselines, even when more data is available per task. This highlights that, under large distribution shifts, meta-learning delivers a consistently competitive performance floor, largely independent of sample size. Interestingly, the FT approach does not reach the performance level of the individual baseline model despite the increase in data, reflecting the difficulty of FT when heterogeneity is high. Prototypical Networks perform best at low-to-moderate heterogeneity, where interpolating target values in the latent feature space is appropriate for similar tasks; however, under extreme variability, MAML — with its rapid adaptation mechanism — demonstrates superior flexibility and effectiveness in highly dynamic environments.

As a practical takeaway, we can see how meta-learning offers a robust default strategy, delivering reliable results regardless of task heterogeneity or the amount of adaptation data. FT, by contrast, is opportunistic: it can match meta-learning performance, but only when the new tasks are sufficiently homogeneous and when a large pool of labeled support instances is available.

4.3. Extreme few-shot

Extreme few-shot refers to a scenario that is even more demanding than standard few-shot configurations, in which the number of labeled instances available for training is extremely limited. In certain industrial contexts, obtaining such labels can be costly or time-consuming. Therefore, this section specifically addresses this issue.

Table 4 reports R^2 under extreme few-shot settings (5 and 10 support instances) and a higher-support baseline (25-shot). Meta-learning methods account for most top ranks, with Prototypical Networks and MAML recurrently leading, whereas Fine-tuning tends to lag under higher task heterogeneity. SNAIL shows pronounced variability at very low support, including negative R^2 values; recall that $R^2<0$ indicates performance below a mean-predictor baseline and, in this context, signals instability. Baselines follow expected trends and remain secondary to meta-learning: the Individual model struggles under data scarcity, and SimCLR-based variants benefit modestly from representation pretraining as support increases.

A clear trend emerges at 5-shot: performance decreases as task heterogeneity increases, dropping from Mango (0.601 with MAML) to TRIP (−0.095 with MAML). This is consistent with the intuition that, in regression, five support examples rarely cover the target range under distribution shift. Increasing support to 10–25 examples substantially stabilizes adaptation, particularly for metric-based meta-learning, which benefits from more reliable prototype estimation. These observations are consistent with the benchmark-wide pattern reported in this work.

As our 5-shot support is randomly sampled, it rarely spans the target domain in regression, particularly under high heterogeneity, whereas 10–25 shots provide markedly better coverage and stability. Therefore, although meta-learning remains competitive in extremely limited scenarios, careful support selection that considers diversity is crucial in heterogeneous settings.

4.4. Meta-learning performance analysis under increased task availability

In addition to evaluating the effect of data availability per task, we investigated how increasing the number of *meta-training tasks* influences model performance. Specifically, we focused on the TRIP test, which was previously identified as the most heterogeneous and challenging scenario (Section 3.3). This setting allows us to examine the ability

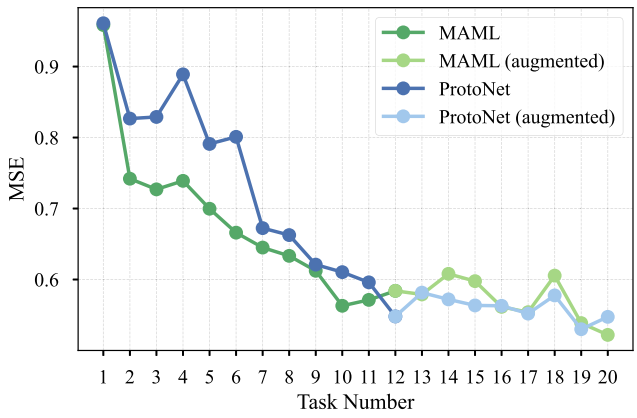


Fig. 3. Mean Squared Error (MSE) of MAML and Prototypical Networks as a function of the number of meta-training tasks. The "augmented" variants incorporate 9 additional tasks (numbered 12 to 20) from unrelated datasets (Mango, Soil NIR, and Soil MIR), demonstrating that performance can continue to improve even with highly heterogeneous training tasks.

of meta-learning methods to generalize across diverse and unrelated prediction problems.

Initially, both MAML and Prototypical Networks were trained on the 11 tasks from the TRIP test. To probe whether performance could be further improved, we progressively introduced 9 additional tasks to the training set, sourced from the Mango, Soil NIR, and Soil MIR tests. These tasks originate from completely different distributions and do not share characteristics with the original TRIP test, thus representing a substantial distribution shift.

The results, presented in Fig. 3, show that increasing the number of meta-training tasks leads to a consistent improvement in performance for both models. Importantly, this trend persists even when the new tasks are highly heterogeneous and unrelated to the target tasks. The augmented models (those trained on all 20 tasks) outperform their counterparts trained only on the original 11 TRIP tasks, achieving lower MSE across the held-out test tasks.

These findings suggest that meta-learning algorithms are capable of extracting transferable knowledge even from task distributions that differ significantly. This has important implications for the development of general-purpose, foundational models for spectroscopy. By aggregating a sufficiently broad and diverse set of tasks, it may be possible to build models that generalize robustly to new scenarios without requiring extensive adaptation data.

4.5. Latent-space structure analysis

This subsection investigates how latent spaces behave under two adaptation strategies — Prototypical Networks, the best-performing meta-learning approach in our study, and FT — to elucidate their representational differences. To cover contrasting regimes, we analyze a highly heterogeneous setting from the TRIP benchmark and a more homogeneous setting from Mango.

For TRIP, three tasks were selected so that they share exactly the same support and query spectra (CGL tasks: casein, lactate, glucose); this isolates the effect of changing the supervised target while keeping the input distribution fixed. For Mango, one representative task was analyzed. To emulate few-shot adaptation while ensuring minimally stable estimation, 25 support instances per task were used. Prototypical Networks operate with a shared encoder and do not re-adapt the backbone at test time, relying on prototype comparisons in the learned metric space; in contrast, FT adapts the backbone independently for each task using the same 25 support spectra.

For visualization, penultimate-layer embeddings were extracted — after adaptation for FT, and from the shared encoder for Prototypical Networks — and projected to two dimensions via PCA. All available query instances were included for each task (note that the query set is identical across the three CGL tasks). As a quantitative companion to the plots, we report a target-latent alignment score $r = \max\left(|\mathrm{corr}(y, \mathrm{PC1})|, |\mathrm{corr}(y, \mathrm{PC2})|\right)$, which indicates the presence of a near-monotonic one-dimensional manifold aligned with the target.

The resulting two-dimensional projections are shown in Figs. 4–5. In the heterogeneous TRIP subset, *Prototypical Networks* learn a shared encoder and metric space where distinct properties (casein, lactate, glucose) occupy well-separated regions organized along smooth, almost monotonic gradients of the target. Crucially, this geometry emerges *without re-adapting* the encoder at test time, relying only on prototype comparisons. By contrast, FT produces task-specific embeddings that frequently compress variability or mix target levels, yielding weaker target–latent alignment and fragmented manifolds. The visual patterns align with the quantitative gap previously reported in Table 3: when support is scarce and inter-task shift is high, the metric inductive bias of *Prototypical Networks* better preserves cross-task invariances, whereas FT is prone to overfitting idiosyncrasies and fails to recover a representation that separates multiple properties reliably.

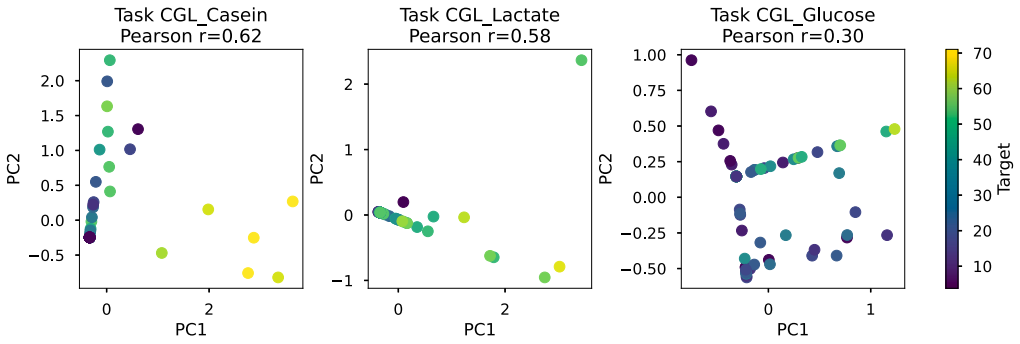
The Mango subset exhibits substantially higher cross-task homogeneity, and all tasks address the same target (dry matter). In this regime the encoder does not need to support several properties simultaneously; a nearly one-dimensional monotonic manifold suffices. Accordingly, *Prototypical Networks* and FT induce very similar latent organizations with consistently high alignment ($r \approx 0.93$–0.94). This mirrors the learning curves previously shown in Fig. 2, where FT approaches the meta-learning performance as a few additional support samples become available. When the supervised objective is shared and distributional shift is small, the extra metric structure imposed by *Prototypical Networks* has limited room to improve over adapting a well pre-trained encoder via FT.

4.6. Robustness to spectral noise and instrument drift

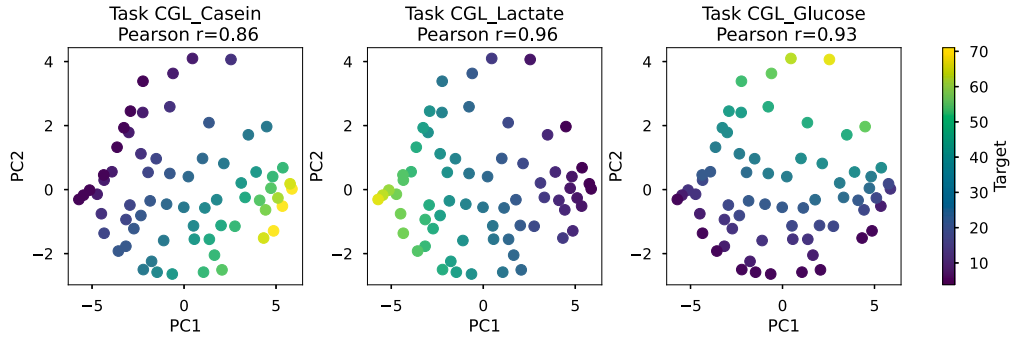
Although all of the data sets used come from real industrial environments and therefore already incorporate typical noise and variations, we performed a stress test to check whether the model would maintain relatively robust behavior in the face of more aggressive noise disturbances. Three families were considered during meta-training and adaptation that are typically found on spectrometry machinery (Rinnan et al., 2009; Roger et al., 2022): (i) additive Gaussian noise (AWGN, or white noise with a normal distribution added to the signal), (ii) instrumental drift (smooth changes in gain and/or small shifts in the spectral axis), and (iii) baseline noise (low-frequency trends), as well as a combination of these. As expected, injecting disturbances decreases predictive performance; noise typically harms ML models. However, in our experiments, the deterioration was moderate, and the overall trends were preserved, indicating the approach's practical robustness. Fig. 6 shows the mean squared error (MSE) curves as a function of the number of support instances in the three tests (Mango, Soil, and TRIP). The patterns observed are consistent with the original clean case, and the relative behavior between tests does not vary, although the degradation is more pronounced in Mango. Interestingly, under extreme few-shot and high heterogeneity (TRIP), moderate noise injection behaves like data augmentation and slightly improves performance; in that regime, the combination of the three perturbations (AWGN + drift + baseline) outperforms any of the individual noise types.

4.7. Qualitative interpretability case study (Mango)

To illustrate how explainability can be layered on top of our models, we show wavelength-level attributions for a representative Mango test instance. Fig. 7 presents the original spectrum together with attribution profiles obtained with SHAP and XGrad-CAM. SHAP provides additive, input-level contributions per wavelength, whereas XGrad-CAM localizes influential contiguous regions in the encoder's activation space. We report these plots as qualitative, model-derived indicators intended to assist expert assessment rather than definitive band assignments.



(a) *Fine-tuning*. Task-specific embeddings after adaptation; compressed layouts and weaker target–latent alignment (moderate r).



(b) *Prototypical Networks*. Shared embedding across targets (no encoder adaptation); clear monotonic target gradients and high r.

Fig. 4. PCA of penultimate-layer embeddings for three tasks from the CGL subset of TRIP (casein, lactate, glucose), computed on the same 67 spectra. Color encodes the target value. Each panel reports the Pearson alignment $r = \max(|r_{PC1}|, |r_{PC2}|)$. Prototypical Networks (bottom) display smooth, monotonic target gradients and higher alignment than fine-tuning (top).

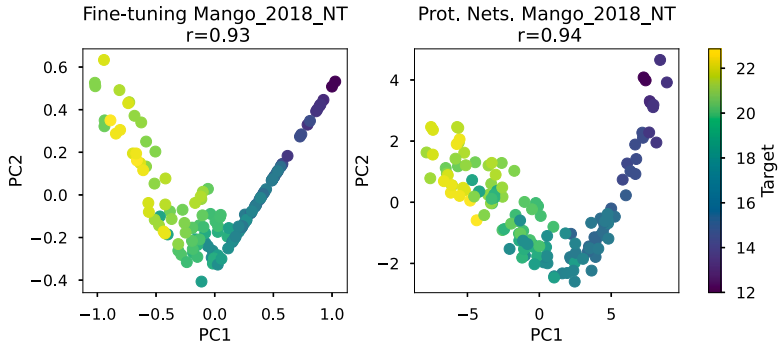


Fig. 5. PCA of penultimate-layer embeddings for *Mango_2018_NT*. Color encodes the target value. Both fine-tuning (left) and Prototypical Networks (right) yield very high Pearson alignment ($r \approx 0.93$–0.94), indicating a nearly one-dimensional monotonic manifold and a reduced gap compared to the heterogeneous TRIP tasks.

5. Discussion

In this work, we propose a methodological workflow that aims to address spectrometric regression problems in industrial settings under few-shot conditions. The initial step involves the quantification of task heterogeneity, for example, through the utilization of distribution distance metrics or feature-importance measures. In scenarios with low heterogeneity and a sufficient number of samples, the FT strategy proves to be the most straightforward and effective approach, capitalizing on the inherent similarity between novel tasks and the pretrained model. However, when sample availability is limited — even under low heterogeneity — FT performance degrades, and Prototypical Networks emerge as the best-performing model. As heterogeneity increases, we observe a rapid decline in the performance of simple FT, with only meta-learning methods (such as MAML or Prototypical Networks) maintaining robustness and achieving substantial improvements. These empirical differences are mirrored in the latent-space visualizations, where Prototypical Networks produce smooth, target-ordered embeddings that ease adaptation to new tasks. Notably, despite being a meta-learning method, SNAIL exhibits poor performance and fails to successfully address most of the proposed regression scenarios. More specifically, some studies, such as Finn and Levine (2017), have shown that black-box meta-learners, particularly SNAIL, are less robust to distribution shifts and high variability between tasks than metric- or optimization-based methods. These findings aligns with the

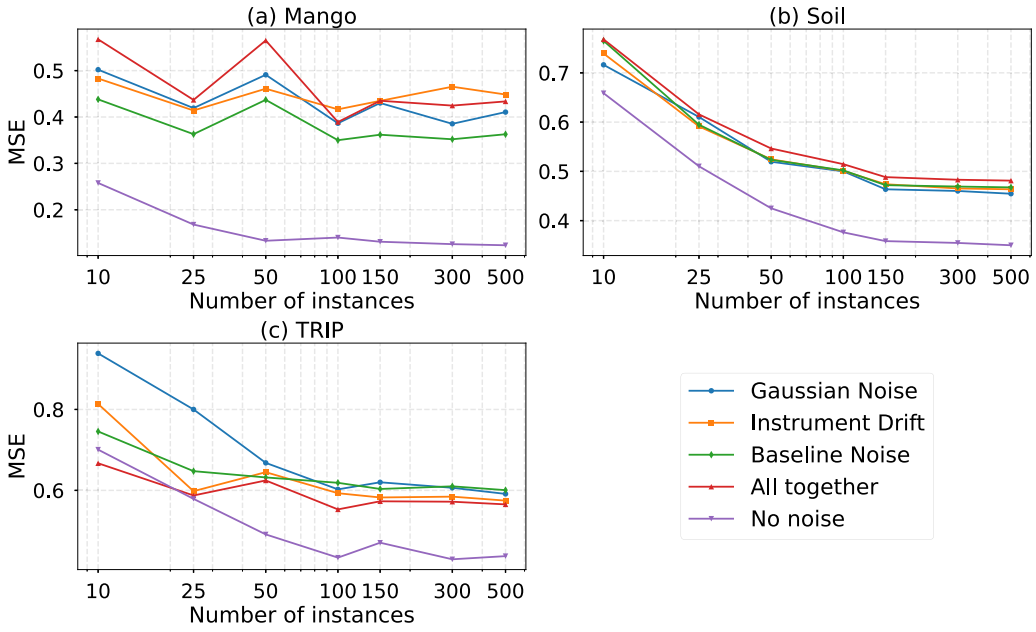
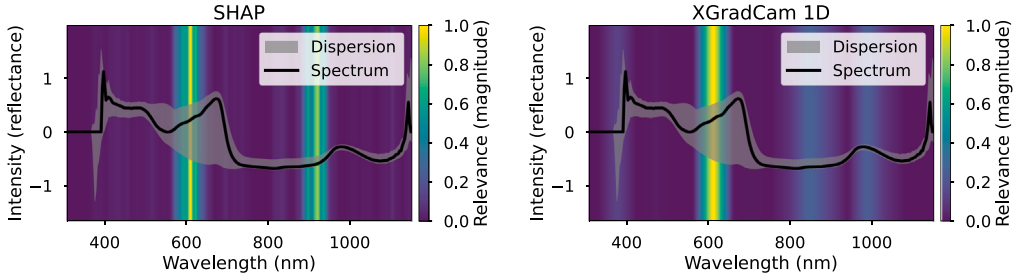


Fig. 6. Robustness of Prototypical Networks to Spectral Disturbances. Mean square error (MSE) is plotted as a function of the number of support instances for Mango (a), Soil (b), and TRIP (c). The plots compare the case without noise to the cases with (i) additive Gaussian noise, (ii) instrument drift, and (iii) baseline noise, as well as the combination of all of them. The curves demonstrate that the overall patterns and error levels remain stable under moderate disturbances, which supports the robustness of the approach in industrial scenarios.



(a) SHAP on a Mango instance.

(b) XGrad-CAM (1D) on the same instance.

Fig. 7. Qualitative XAI overlays for the Mango test. SHAP offers input-level, additive attributions per wavelength, while XGrad-CAM localizes contiguous spectral windows at the layer level. These visualizations are intended to support expert auditing and domain-specific assessment. In regulated applications, the combination of additive and localization views enhances transparency and audit readiness. Notably, the strongest attribution bands in both panels align with the highest-dispersion zones of the spectra (gray), suggesting these windows harbor the predictive differences across samples.

results in Section 4, where greater task heterogeneity results in lower performance.

Moreover, our results demonstrate that the TRIP dataset can serve as a "universal" starting point in industrial applications. Even for highly disparate problems, using the *TRIP test* tasks for meta-training produces better outcomes than training from scratch or applying FT directly. This suggests that, in practice, maintaining a diverse spectral bank facilitates rapid adaptation to new sites or sensors without requiring the collection of large volumes of task-specific data.

To improve transparency without changing model design, we additionally integrate lightweight, model-agnostic explainability overlays — SHAP (GradientExplainer) and XGrad-CAM — that provide complementary input-level and layer-level views of wavelength relevance. These qualitative visualizations are intended to assist expert assessment rather than to assert physico-chemical assignments, and they complement quantitative metrics in few-shot regimes.

Despite these encouraging results, two significant limitations require attention. First, meta-learning methods such as MAML incur significant computational overhead due to second-order gradient calculations, which limits their applicability in resource-constrained settings if meta-training is necessary. However, once the model is meta-trained, the adaptation process is exactly the same as standard fine-tuning, eliminating the need for additional costs. Secondly, hyperparameter sensitivity is not uniform across methods; MAML is more sensitive than Prototypical Networks and typically requires more careful tuning to achieve stable performance (Appendix B.3).

Finally, we emphasize the need to build a foundational spectroscopy dataset at a substantially larger scale, encompassing multiple domains, and sampling conditions. A foundational model trained on such a corpus would enable industrial applications with minimal data-collection costs, accelerating deployments and reducing calibration errors.

6. Conclusions

In this work, we introduced a unified benchmark that consolidates multiple spectral libraries from diverse industrial applications into a single, task-oriented resource. This structured organization enabled a rigorous and transparent evaluation of various machine learning approaches, including deep learning, transfer learning, and notably, meta-learning techniques.

Our experimental results demonstrate that meta-learning methods unequivocally achieve the highest performance across all few-shot spectroscopic regression tasks, consistently outperforming both FT strategies and individually trained models. While FT can occasionally match meta-learning under highly homogeneous conditions with ample labeled data, it remains opportunistic and sensitive to task similarity. In contrast, meta-learning yields superior accuracy irrespective of support-set size or inter-task heterogeneity. This effect is most striking in the TRIP benchmark — our most heterogeneous evaluation — where meta-learning algorithms such as MAML and, notably, Prototypical Networks improve performance by over 80% relative to FT. Consistent with the latent-space analysis, these gains are supported by a more structured embedding geometry — shared, target-ordered manifolds with clearer cross-property separation — which facilitates prototype-based adaptation under distribution shift. To our knowledge, this study constitutes the first comprehensive demonstration of meta-learning's effectiveness in spectroscopic analysis, firmly establishing its potential as a robust default approach for rapid adaptation in industrial settings. Furthermore, the application of task heterogeneity metrics such as MMD and FIS allowed us to objectively quantify the variability among tasks, confirming that meta-learning approaches provide significant benefits when facing large distributional differences.

We also provide a practical methodology. First, quantify task heterogeneity. If heterogeneity is low and ample data is available, FT or meta-learning can yield similar results; as heterogeneity increases or data becomes scarce, switch to meta-learning. Whenever possible, pre-train on the TRIP set to accelerate adaptation to new tasks. This roadmap can guide practitioners in industrial settings to achieve strong performance even with very limited data. As an additional contribution of this work, the benchmark code and preprocessing pipelines have been made publicly available.

Building on the qualitative XAI analyses presented in Section 4.7, future work should standardize and quantitatively validate XAI within the benchmark—e.g., defining domain-validated band checks for auditability in regulated settings. Additionally, it would be beneficial to maintain and continuously update a comprehensive spectroscopic data repository to progressively expand the TRIP dataset; this would not only enrich sample diversity but also facilitate the derivation of more robust meta-trained weights for specific industrial applications. Furthermore, it is essential to validate the proposed methodology on new datasets — both NIR/MIR spectra and other data sources — to confirm the generality and robustness of the obtained conclusions while identifying potential biases or limitations across different scenarios. In future works, it would be interesting to explore other spectroscopy-specific architectures with inductive biases tailored to 1D spectra. Finally, to broaden the benchmark's scope, introducing spectra from other wavelength modalities (e.g., UV–Vis, Raman, or terahertz) would permit evaluation of whether meta-trained weights derived from NIR/MIR spectra can be efficiently transferred or adapted to these modalities, thereby offering more versatile solutions for spectral characterization challenges in highly heterogeneous environments.

CRediT authorship contribution statement

Iván Garzón: Writing – review & editing, Writing – original draft, Visualization, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Guillermo Gomez-Trenado:** Writing – review & editing, Validation, Supervision, Methodology, Investigation, Conceptualization. **Sergio Damas:** Writing – review & editing, Validation, Supervision, Project administration, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used *ChatGPT (OpenAI)* in order to refine the language, improve coherence and clarity of the manuscript. After using this tool/service, the author(s) reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.engappai.2025.113593.

Data availability

We release the full benchmark pipeline and automated scripts to obtain and prepare each dataset under its original terms; we do *not* mirror third-party data. **109,858 of 111,767 instances (98.29%)** come from **open-licensed sources** (CC BY/MIT), which generally allow reuse — including *commercial* use — with proper attribution, whereas **1909 instances (1.71%)** correspond to sources whose landing pages do *not* state an explicit license. Although datasets in this latter group have been previously used in the spectroscopy/chemometrics literature, they should not be redistributed and *commercial* use requires authorization. For users operating under commercial constraints, our repository offers a convenient option to **download only the datasets whose licenses allow commercial use**.

The full benchmark pipeline, preprocessing scripts, and experimental code are publicly available at: https://github.com/ivgase/spectroscopy-fewshot-benchmark.

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